

Critical Slowing Down: Financial Crisis Prediction

Early Warning Signals in Financial Markets

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- **Research Question:** Can we detect early warning signals of financial crises using Critical Slowing Down (CSD)?
- **Motivation:** Predict potential financial transitions before they occur
- **Interdisciplinary Approach:** Applying physics and complexity theory to financial markets

Overview

1. Critical Transition Theory
2. Data
3. Methodology
4. Black Monday Analysis
5. 2008 Financial Crisis
6. Conclusion

1. Critical Transitions Theory

Key Concepts:

- Bifurcation: Abrupt change in system state
- As a system approaches a critical transition, it becomes slower to recover from disturbances

Critical Slowing Down Indicators:

- AR(1) Autocorrelation
- Variance
- Skewness

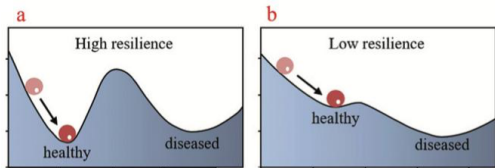


Figure: Illustration of a Critical Transition

2. Data

Crisis	Critical Point	Index
Black Monday	13 Oct 1987	S&P 500
2008 Financial Crisis	15 Sept 2008	S&P 500, TED Spread, FTSE 100

Table: Financial Indices and Critical Points

3. Methodology

Detrending Process:

- Gaussian Kernel Smoothing to remove long-term trends
- To get rid of patterns

Effect of σ :

- Small σ : Focus on nearby points
- Large σ : Include distant points

Parameters:

- Fixed total window: 800 observations
- Window sizes: $T/2$, $T/4$, Robustness test

Robustness Test:

- σ : 5 to 20
- Window sizes: 50 to 200

4. Black Monday - Analysis of AR(1)

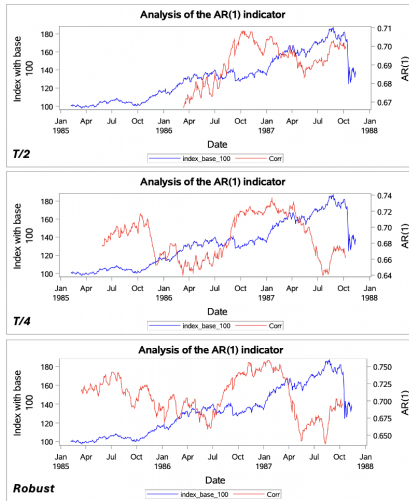


Figure: Evolution of the AR(1) for each window for Black Monday

4. Black Monday Analysis - Analysis of Variance

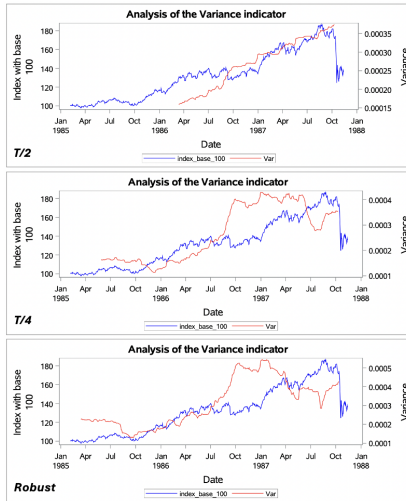


Figure: Evolution of the variance for each window for Black Monday

4. Black Monday Analysis - Analysis of Skewness

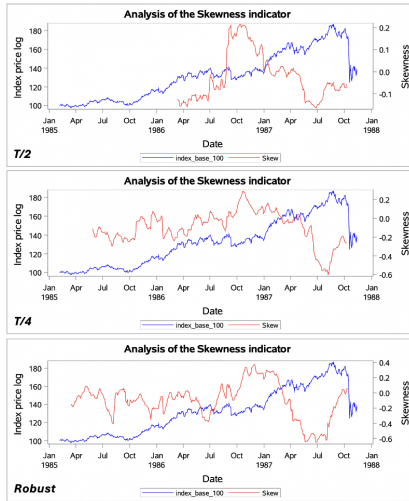


Figure: Evolution of the skewness for each window for Black Monday

5. 2008 Financial Crisis - Analysis of $Ar(1)$ - S&P500

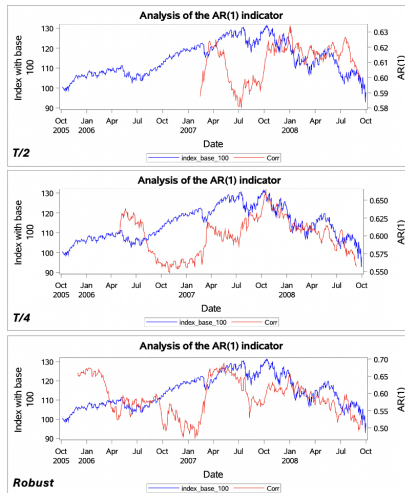


Figure: Evolution of the $Ar(1)$ for each window for 2008 Financial Crisis

5. 2008 Financial Crisis - Analysis of Variance - S&P500

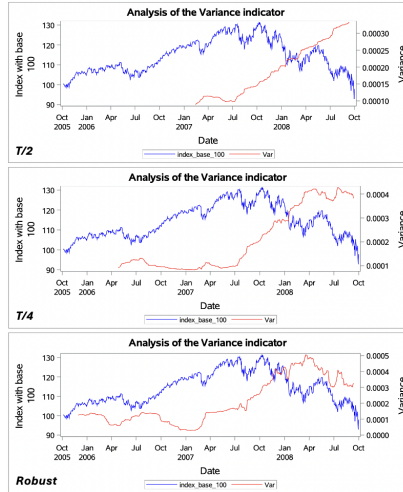


Figure: Evolution of the Variance for each window for 2008 Financial Crisis

5. 2008 Financial Crisis - Analysis of Skewness - S&P500

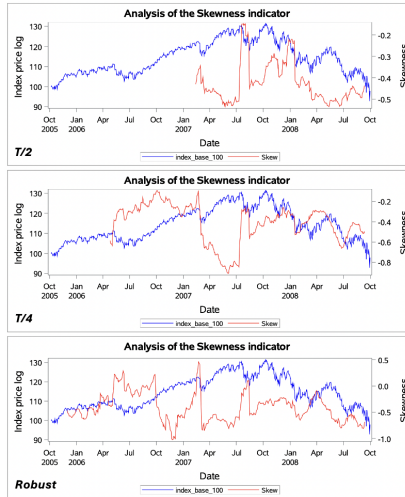


Figure: Evolution of the Skewness for each window for 2008 Financial Crisis

6. Conclusion

Key Findings:

- CSD promising for crisis prediction
- Variance most reliable indicator
- AR(1) results complex and nuanced

Overall Assessment

Critical Slowing Down: A versatile but effectiveness varies by crisis type

**THANK YOU FOR LISTENING
TO US**